High-Resolution Density Joint Inversion Method of Airborne and Ground Gravity Data With Cross-Constraint Technique

Qingfa Meng, Guoqing Ma, Taihan Wang, and Tingyi Wang

***Abstract*— The airborne gravimetry measurement can better highlight the anomalous response of deeper resources than ground gravity data, so an integrated survey of airborne and ground gravity data is an effective strategy to obtain the obvious response of shallow and deep mineral resources simultaneously. In order to obtain a more reliable and higher resolution 3-D density distribution model through the joint inversion of airborne and ground gravity data to support deeper source exploration, we propose a high-resolution joint inversion method with a cross-constraint technique. We use structural constraints to establish an objective function for joint inversion of airborne and surface gravity data, and add density-weighting constraints through cross-calculation to make full use of the sensitivity of different data and geological targets. Synthetic model tests prove that the cross-constraint joint inversion method of airborne and ground data improves the model resolution effectively, and can delineate the distribution of deeper targets more clearly compared to the existing joint inversion way. In order to predict the spatial distribution of deep ore-bearing rock bodies in a north Qinling area of Shaanxi province, we obtained regional measured airborne and surface gravity data. High-resolution density joint cross-constraint inversion method was used for real data interpretation, and the distribution feature of possible deep ore-forming rock bodies.**

***Index Terms*— Airborne and ground gravity data, cross constraint, high resolution, joint inversion.**

I. Introduction

AIRBORNE gravity measurement is widely used in the exploration of mineral resources because of its high efficiency and strong practicality [1]. Compared to ground gravity data, the airborne gravity measurement can highlight the anomalies of deep geological bodies, so the joint exploration of airborne and ground gravity data can obtain the distribution feature of shallow and deep sources simultaneously [2], [3]. The joint inversion of airborne and ground gravity data is widely used to get accurate source location parameters [4]–[6]. Fedi [7] presented the depth from extreme points (DEXP) method, which used the extreme points of the data with different heights to obtain the location of the source. Zhou *et al.* [8] used the Euler deconvolution of gravity data with multiple heights to get convergent results.

The 3-D density inversion is an important tool in the interpretation of gravity data [9]–[20]. We all know that the inversion is unascertained, and the joint inversion of multiple data can reduce the algebraic ambiguity to get more accurate inversion results [21], [22]. Zhang and Zhang [23] proposed the equivalent-dimensions density inversion method of gravity data with different heights, which combined data with different heights into a data matrix, and can get a more accurate density distribution. Fedi *et al.* [4] and Wang *et al.* [24] also used the joint inversion of airborne and ground data by strategies that combine all the data into a matrix to obtain the distribution of geological bodies. These studies show that the joint inversion of gravity data with multiple heights would provide more accurate results compared to the inversion results computed by single-height gravity data. However, the inversion resolution of the strategy that involves airborne and ground gravity data into a matrix is not improved obviously.

To improve the resolution of the joint inversion of airborne and ground data for deeper source exploration, we propose a high-resolution inversion method with cross-constraint technique (abbreviated as cross-constraint joint inversion method), which uses a multiconstrained manner of data and structure to obtain higher resolution density results compared with conventional computation way. The synthetic tests and real data show that the cross-constraint joint inversion method can obtain higher resolution density results, and can give a distribution of deep sources more clearly.

II. Cross-Constraint Joint Inversion Method

The 3-D density inversion usually uses the way that discretizes subsurface into a prism, and then computes the density of each prism. The current strategy of the joint inversion of airborne and ground gravity data is that combine gravity data with different heights into a matrix (abbreviated as combined gravity) to compute the density distribution [4],

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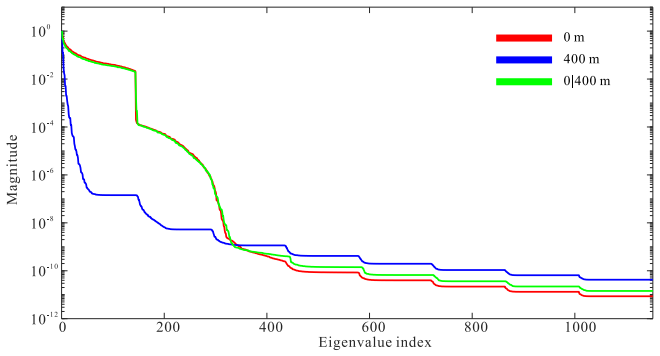


Fig. 1. Eigenvalue spectra for kernel function matrices of different and multiple observation heights gravity. Spectra are normalized with respect to the largest eigenvalue.

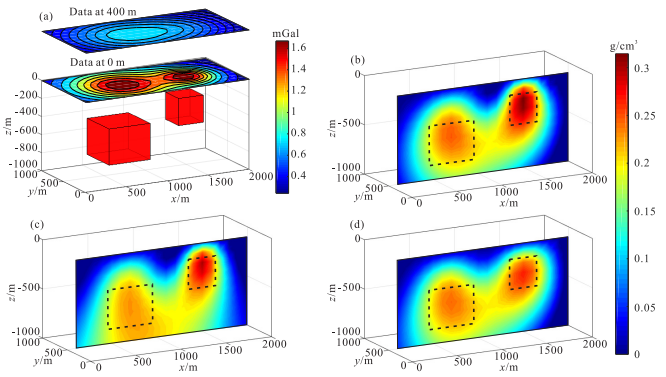


Fig. 2. Regularization inversion results of combined, ground, and airborne gravity data. (a) Airborne and ground gravity and the designed density models. (b) Slice of model from combined gravity inversion result. (c) and (d) Slices of model from ground and airborne gravity inversion result.

and the corresponding objective function of the regularization inversion is

$$\begin{aligned}\phi &= \phi_d + \mu\phi_m \\ &= \left\| W_d \left(\begin{bmatrix} G_g \\ G_a \end{bmatrix} m - \begin{bmatrix} d_g \\ d_a \end{bmatrix} \right) \right\|_2^2 + \mu \| W_m \|_2^2 \to \min \end{aligned} \quad (1)$$

where ϕ_d and ϕ_m are the function of data fitting and model stabilizing, respectively, W_m is the selected stabilizer which is the diagonal matrices of m. W_d is the depth weighting function, and following Portniaguine and Zhdanov, we apply the weighting function in the data fitting function [12], [13] and μ is the regularization factor to control the level of model stability. d_g and G_g represent the ground gravity data and its kernel function matrix, and d_a and G_a represent the airborne gravity data and its kernel function matrix, and m is the density of prisms. The kernel function matrixes G_g and G_a are computed by the forward formula of prism [25].

The larger magnitude of eigenvalue spectrum indicates the higher relative resolution, and the larger eigenvalue index is used to calculate deeper area [26]. The normalized eigenvalue spectrum can be used to analyze the relative resolution of different observation heights gravity data at different depths. The calculated model is a 3-D volume containing 1152 ($12 \times 12 \times 8$) prism cells with different observed heights. We use the data at 400 and 0 m to represent airborne gravity and surface data, and the eigenvalue spectra are shown in Fig. 1.

We find that the combined gravity of 0 and 400 m has the same magnitude as the surface data in a small eigenvalue index range, and has a higher magnitude than the surface data in a large eigenvalue index range. So, the joint inversion strategy that combined gravity data with different heights into a matrix can better get the distribution of the deep geological body than the inversion results by single surface gravity. The data combination is higher than the single airborne gravity data in a small eigenvalue index range, but it is lower in a large eigenvalue index range, which indicates that joint inversion can obtain shallow source location parameters more precisely.

We design a model shown in Fig. 2 to verify the above conclusions from the eigenvalue spectrum. The inversion results slice in Fig. 2(b)–(d) are displayed in the same color range. It can be seen that both the deep model and the shallow model can be identified from the ground gravity anomaly, whereas the airborne gravity data mainly reflect the response of the deep model in the anomalies from Fig. 2(a). The inversion result calculated by the combined gravity data [Fig. 2(b)] has a higher resolution than that obtained by the ground gravity [Fig. 2(c)]. The inversion result of combined gravity inversion result can better recover the distribution of the shallow model than the airborne gravity inversion result [Fig. 2(d)]. However, the inversion of combined gravity data still does not improve the resolution to deep source obviously.

We propose the cross-constraint inversion method to achieve the high-resolution joint inversion for better obtain the distribution of shallow and deep source, which combines the characteristics of the combined gravity and airborne gravity in the inversion through two structural constraints joint inversion, and adds the cross-constraint density weighting function in the twice joint inversion to improve the inconsistency of the resolution caused by the combined and airborne gravity. The specific computation process of the cross-constraint joint inversion method is shown in Fig. 3.

The cross-gradient is calculated as the structural constraint for joint inversion of combined gravity and airborne gravity data. It can combine the data constraint with multiple depths and the structural constraint in the operation to get higher resolution inversion results (m_{a1} and m_{c1} in Fig. 3). The cross-gradient function [27], [28] can be written as

$$\vec{t}(x, y, z) = \nabla m_1(x, y, z) \times \nabla m_2(x, y, z) \quad (2)$$

where m_1 and m_2 represent two sets of parameters in three-dimensional space, which are in our case the density models estimated from combined and airborne gravity data. The computation of cross-gradient is carried out in the x-, y-, and z-directions, and the specific expressions are

$$\begin{aligned} t_x &= \frac{\partial m_1}{\partial y}\frac{\partial m_2}{\partial z} - \frac{\partial m_1}{\partial z}\frac{\partial m_2}{\partial y}, \\ t_y &= \frac{\partial m_1}{\partial z}\frac{\partial m_2}{\partial x} - \frac{\partial m_1}{\partial x}\frac{\partial m_2}{\partial z}, \\ t_z &= \frac{\partial m_1}{\partial x}\frac{\partial m_2}{\partial y} - \frac{\partial m_1}{\partial y}\frac{\partial m_2}{\partial x}. \end{aligned} \quad (3)$$

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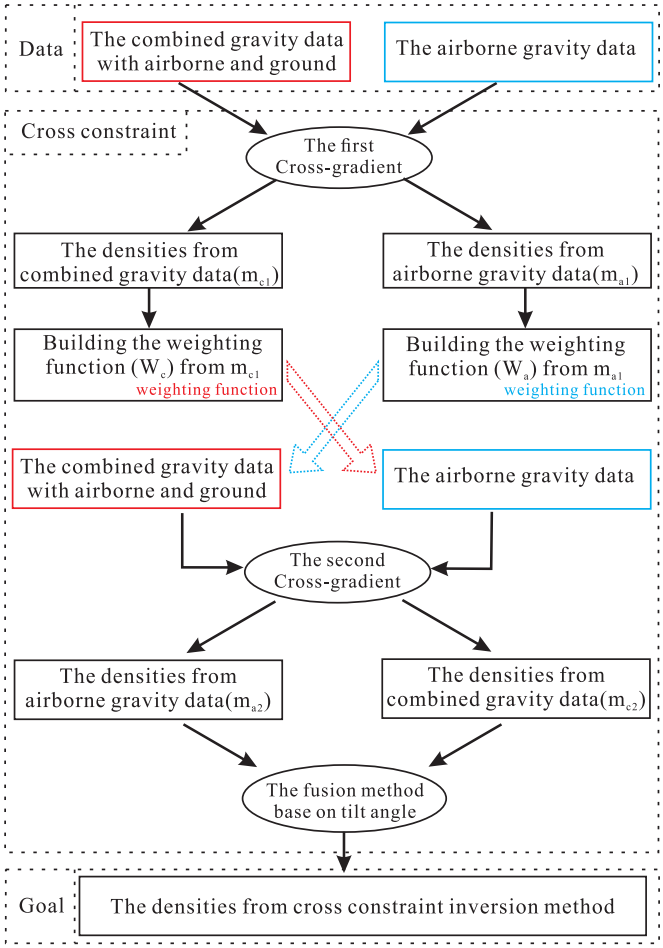


Fig. 3. Flowchart of cross-constraint joint inversion.

We add the cross-gradient term to the objective function as:

$$\phi_{\text{cross}} = \|\vec{t}\|_2^2 = t_x^{\text{T}} t_x + t_y^{\text{T}} t_y + t_z^{\text{T}} t_z. \tag{4}$$

The objective function of the cross-gradient joint inversion of combined and airborne gravity data can be rewritten by

$$\begin{aligned}\phi &= \phi_d + \mu\phi_m + \alpha\phi_{\text{cross}} \\ &= \left\| W \begin{bmatrix} \begin{bmatrix} G_a \\ G_g \end{bmatrix} m_c - \begin{bmatrix} d_a \\ d_g \end{bmatrix} \\ G_a m_a - d_a \end{bmatrix} \right\|_2^2 + \mu \|W_m\|_2^2 + \alpha \|\vec{t}\|_2^2 \to \min\end{aligned} \tag{5}$$

where α is a weighting factor to control the strength of structural constraint, and the value is 8–12th power of 10. If this value is too large, the decline of the data fitting degree will be slowed down, and if the value is too small, the constraint strength will be weakened. We selected the μ and α by the trial-and-error method, and the initial values of μ and α are confirmed by making the magnitude of ϕ_d, ϕ_m, and ϕ_{cross} roughly on the same order. m_c and m_a are the densities of inversion results by the combined and airborne gravity data, respectively. In the first cross-gradient inversion progress, W is W_d in (1), and in the second inversion progress, W is the function by adding the density weighting function into W_d.

The density of shallower model from combined gravity inversion is higher than that of deeper model, but the result of airborne gravity inversion is the opposite. (These conclusions will be proved in the first synthetic model.) In order to further improve the vertical resolution of inversion, we introduce the density weighting function into the second cross gradient joint inversion. To be specific, we convert the density of combined gravity inversion into the density weighting function of airborne gravity inversion, and the density of airborne gravity inversion into the density weighting function of combined gravity inversion.

The conversion equation between inversion result and density weighting function is expressed as

$$W_{\text{density}} = (e + m_{\text{norm}})^{-\beta} \tag{6}$$

where, e is a positive constant (less than 10^{-7}) sufficiently close to 0; m_{norm} is the inversion results (m_c or m_a) after normalization; β is the density weighting factor of which value is between 1 and 2. The density weighting function added to W in (5) is the diagonal matrix form of W_{density} of $N \times N$, and N is the number of density results.

After twice the structural constraint joint inversion of combined and airborne gravity, we get two inversion results (m_{a2} and m_{c2} in Fig. 3). However, the inversion results of the method without focused function tend to be lower than the real density of the geological body. We fuse the two inversion results into a more realistic density model through the following operations.

1) The edge of high-density bodies is detected by the second-order tilt angle method [29].
2) Retain the higher density values of two inversion results in the horizontal range of relatively high-value anomalies and the lower values out of these areas.

III. Synthetic Model Tests

We design a model test of two prisms with the same density (1 g/cm^3) and different depths. The forward ground and airborne gravity anomaly are shown in Fig. 2(a). The size and center coordinates of prism models are 500 × 500 × 400 m^3, (650, 450, 600) m and 300 × 300 × 300 m^3, (1450, 450, 350) m, respectively. Regularization joint inversion and joint inversion with cross-constraint way are used to calculate inversion results with the root mean square (rms) error less than 10^{-5} mGal.

Fig. 4(a) and (b) shows the regularization inversion results of combined and airborne gravity data. By comparing Fig. 4(a) and (b), it is clear that the joint inversion results of combined gravity data are more focused than the results inverted by airborne gravity data, and the effect of the shallower model is more obvious. The results show that the inversion results tend to match the gravity response generated by the shallower model, so joint inversion with the constraint of data cannot improve the resolution in the deeper space obviously. Fig. 4(c) and (d) shows the density results (m_{a1} and m_{c1} in Fig. 3) obtained by the first cross-gradient joint inversion of combined and airborne gravity data, and the results highlight the shallow and deep model respectively.

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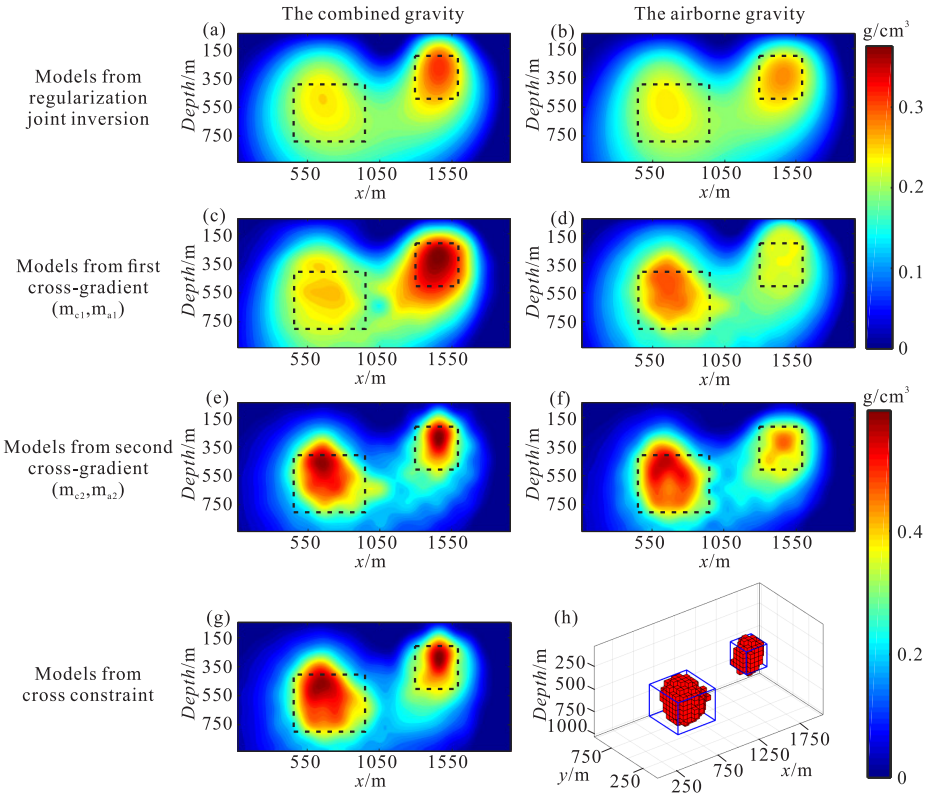


Fig. 4. Models from regularization and cross-constraint joint inversion of ground and airborne gravity. (a)–(g) 2-D slices of density model by inversion. (h) 3-D voxels of density model shown in (g), 2-D slices correspond to $y = 500$ m, and opaque 3-D voxels have densities larger than 0.35 g/cm^3.

Fig. 4(e) and (f) shows the inversion results (m_{a2} and m_{c2} in Fig. 3) after the second cross-gradient joint inversion added density weighting function with the cross-constraint technique. Through comparing these results in Fig. 4(c)–(f), we find the difference of density amplitude between inversion results and that the setting value is smaller in Fig. 4(e) and (f), which provides higher vertical resolution results. As the flow shown in Fig. 3, we fuse these two results in Fig. 4(e) and (f) to get the final inversion results, and it is shown in Fig. 4(g) and (h) by the slice and 3-D voxels. The cross-constraint joint inversion results have higher resolution and more focus, because it combines both data constraint and structure constraint, and it takes the advantage of airborne and ground gravity data.

To verify the universal applicability of the cross-constraint joint inversion method, we forward the gravity at different observation heights with the same models as shown in Fig. 2. Fig. 5(a)–(c) shows the ground and airborne gravity data at 200, 400, and 600 m. The conventional regularized joint inversion results are shown in Fig. 5(d)–(f). In all cases, regularization inversion can obtain the distribution of the shallower inversion results model with the same accuracy, but the deep model of inversion results are not focused enough. The inversion results computed by the cross-constraint joint inversion method are shown in Fig. 5(g)–(i). It is clear that all the cross-constraint joint inversion results show a better distribution than the regularization inversion results. The resolution of the deeper model from cross-constraint joint inversion results has a more accurate density value as the increasing of the airborne gravity observed height, because the gravity of higher observed height can highlight deeper sources. However, higher height of gravity data is not always better, because useful details from the airborne gravity data will be lost.

We use the equivalent source upward continuation method to obtain the gravity data of different heights from the ground gravity data as the alternative to airborne gravity data [30], [31]. For comparing Figs. 5(a)–(c) and 6(a)–(c), the converted gravity data is different from the theoretical data, and the difference increases as the upward continuation height increases. Comparing the conventional regularized inversion results from Figs. 5(d)–(f) and 6(e)–(g), we find that the data conversion error of upward continuation has almost no effect on the inversion result of the shallow model, but it causes a deviation to the recovery of the deep target. Moreover, the area between the shallow and deep models becomes blurred, and it increases as the height increases. Similarly, the position accuracy of the deep model is lower due to the influence of continuation anomaly in cross-constraint joint inversion, and the deviation increases as the height of continuation increases, which are shown in Fig. 6(i)–(k). So, we can conclude that the real gravity can be replaced by the upward continuation gravity when airborne gravity data are observed at low altitudes. We use the anomaly of the model with both positive and negative density bodies to verify the applicability of the cross-constraint joint inversion method. The size and position of the bodies are the same as the bodies in Fig. 5, and the

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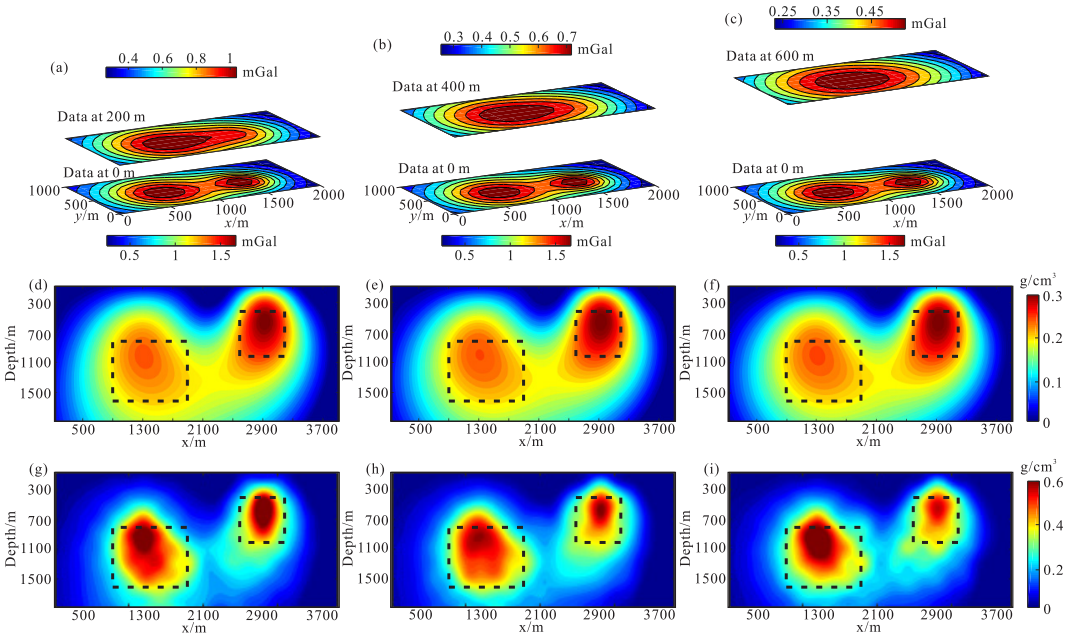


Fig. 5. Models obtained by regularization and cross-constraint joint inversion of ground and different airborne gravity data. (a)–(c) Airborne and ground gravity. (d)–(f) 2-D slices of density model by regularization inversion with combined gravity of (a)–(c). (g)–(i) 2-D slices of density model by cross-constraint joint inversion gravity of (a)–(c). 2-D slices correspond to $y = 500$ m.

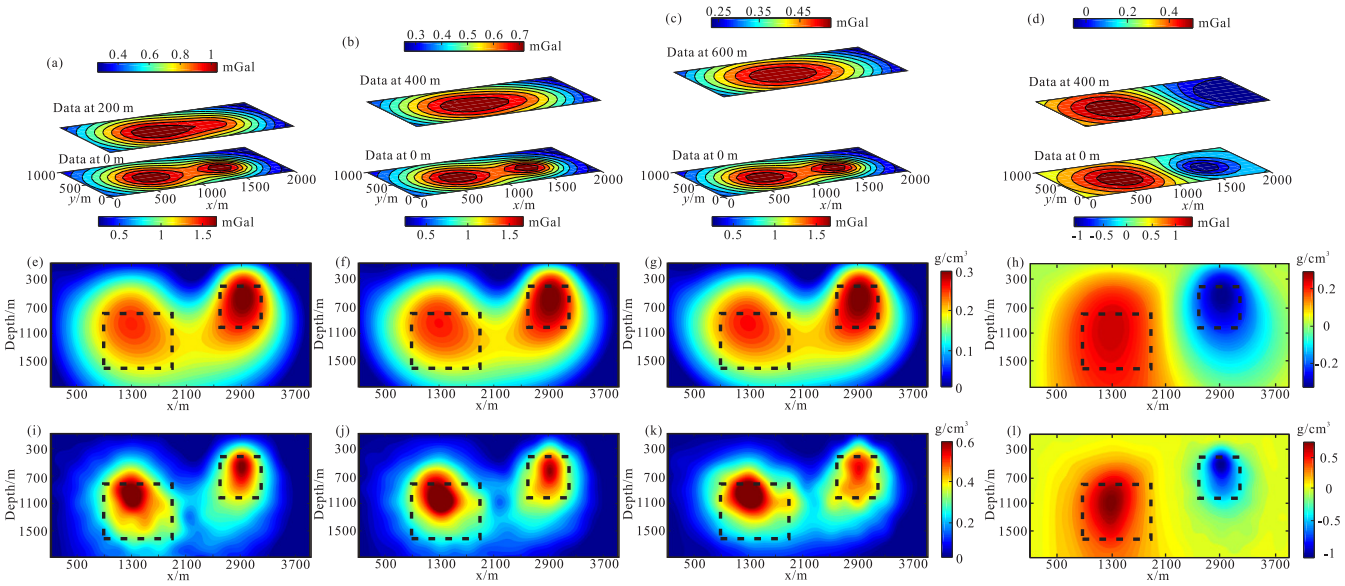


Fig. 6. Models obtained by regularization and cross-constraint joint inversion with transformation and negative gravity. (a)–(c) Upward continuation and ground gravity. (d) Forward gravity with negative anomaly. (e)–(h) Slices of density model by regularization inversion with combined gravity. (i)–(l) Slices of density model by cross-constraint joint inversion, 2-D slices correspond to $y = 500$ m.

residual density of the shallow body is -1 g/cm^3 and the deep body is 1 g/cm^3. Compared to Fig. 6(h) and (l), the cross-constraint joint inversion results are more focused and closer to the real results than the regularization inversion, and the cross-constraint joint inversion method still has the effect of improving inversion resolution.

We designed the synthetic model test with six prisms (1 g/cm^3), and three of them have the same size, which is 300 $\times$ 400 $\times$ 200 m^3, and their center positions are (450, 700, 400), (550, 700, 600), and (650, 700, 800) m, respectively, these prisms are designed to simulate the inclined dike of the actual situation. The other prism sizes are 300 $\times$ 600 $\times$ 300, 300 $\times$ 400 $\times$ 300, and 400 $\times$ 700 $\times$ 800 m^3, and the center positions are (1450, 700, 450), (650, 1500, 350), and (2400, 750, 900) m, respectively; these prisms are designed to simulate the actual situation of high-density ore bodies with different depths and sizes. As shown in Fig. 7, it can be found that the cross-constraint

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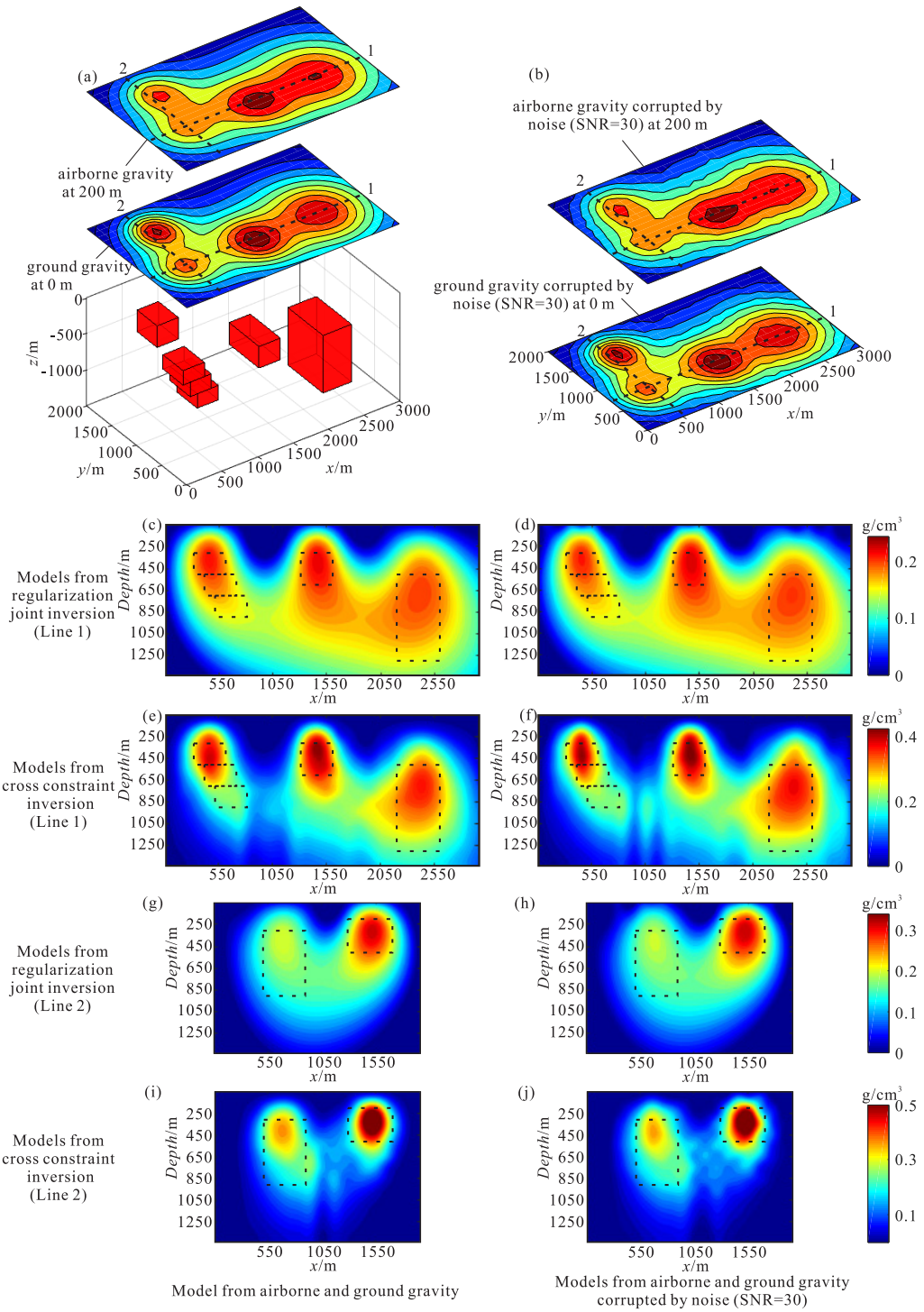


Fig. 7. Synthetic model test of regularization and cross-constraint joint inversion. (a) and (b) Airborne and ground gravity and the designed density models. (c)–(j) 2-D slices of density models by inversion, 2-D slices of line 1 correspond to $y = 750$ m, and 2-D slices of line 2 correspond to $x = 550$ m.

joint inversion method is better than the traditional regularization joint inversion in the case of complex gravity anomalies.

Fig. 7 also shows the gravity anomaly containing 30 SNR noise and the density inversion results. By these results, we can see that the cross-constraint joint inversion method has antinoise performance. Therefore, we can conclude that the result of the cross-constraint joint inversion method has higher resolution and accuracy compared with the traditional regularization inversion result in the complex model test.

IV. Actual Gravity Data Processing

The advantages of airborne and surface joint inversion have been demonstrated, so we made the airborne and surface gravity surveys in the Shaanxi mining area to explore the

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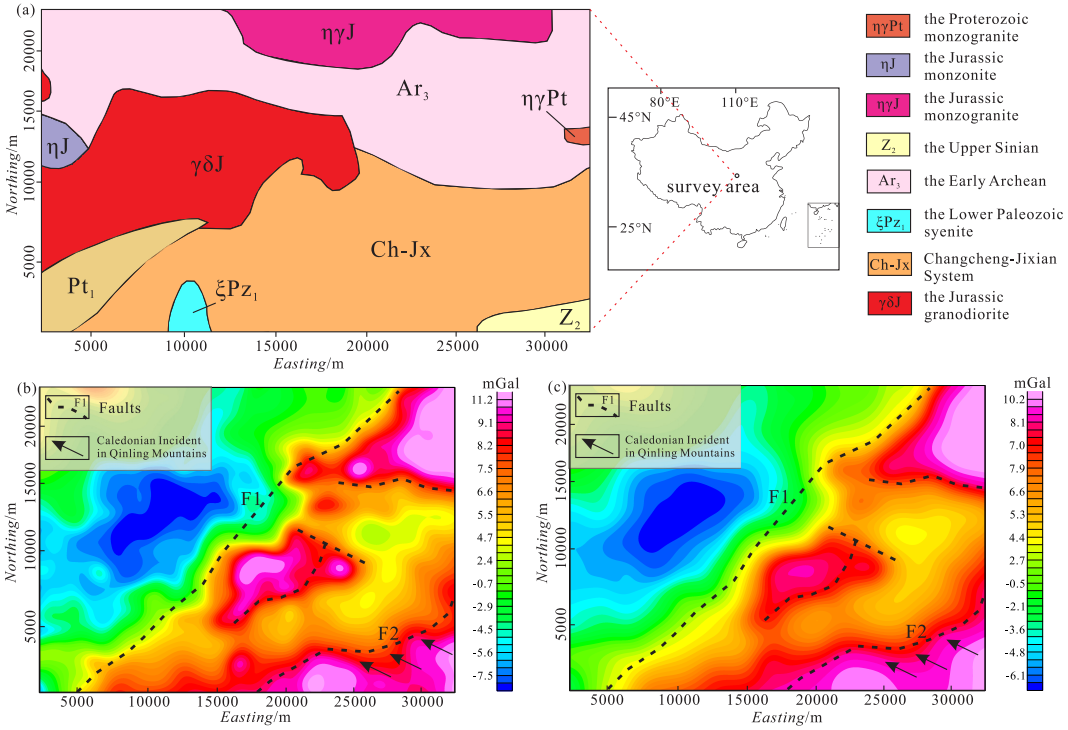


Fig. 8. Geological and Bouguer gravity anomaly maps. (a) Geological map. (b) Ground gravity anomaly. (c) Airborne gravity anomaly in 700 m height.

distribution of high-density ore bodies. We use the cross-constraint joint inversion method to interpret the airborne and surface gravity data of the metal mines' distribution. The inversion parameters μ and α in (5) are 0.01 and 10^{10} in this real data inversion. Fig. 8(a) shows the location and geological information of the study area. The outcrops in this region are mainly monzonite, syenite, granodiorite, and monzogranite. There are mainly polymetallic ore bodies in this area, and magma intruded into the shallow part along the fault to become intrusive rocks through multiple periods of tectonic movement. Then, intrusive rocks become polymetallic ore that existed in different surrounding rocks which show a characteristic of high density. Fig. 8(b) and (c) shows the faults based on geological data [32], and these faults were caused by the Caledonian Incident in the Qinling Mountains of a northwest tectonic stress on this region. To obtain the accurate range of the shallow and deep bodies simultaneously, we use airborne and ground gravity exploration to analyze the distribution of mines. Fig. 8(b) and (c) shows the airborne and surface gravity data, and the airborne gravity has been reduced to 700 m horizontal plane by frequency domain method [33], and the interval space is 500 m. It can be seen that there is a certain correlation between geological and gravity maps. The high-value anomaly concentrates on the border between the Changcheng–Jixian system and Jurassic granodiorite. We finally concluded that the high-density body is the result of magmatic intrusion which is a favorable area of mineralization.

Figure 9 shows the airborne and ground gravity anomalies, and the density inversion results computed by cross-constraint joint inversion methods and the locations of the slices are labeled by black dotted lines (Fig. 9).

Depending on the inversion results, we presume the center depths of high-density bodies are concentrated from

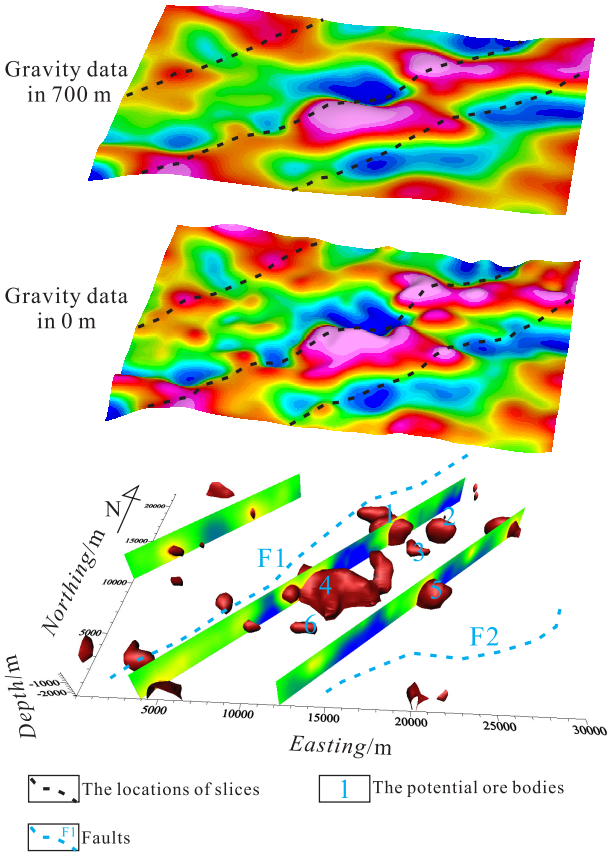


Fig. 9. 3-D map of airborne and ground inversion results computed by cross-constraint joint inversion method, 3-D results greater than 0.12 g/cm^3.

800 to 1600 m. The research area is dominated by magmatic mineralization, based on geological information, so we presume the high-density bodies close to the faults formed by

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magma intrusion through the natural channel of faults. The high-density bodies with complete gravity anomaly between the F1 and F2 are the main potential high-density ore bodies. There are six high-density bodies in this area, and as shown in Fig. 9 with numbers 1–6. The geological bodies of numbers 3 and 6 and the east of number 4 are the shallowest in this area with a depth of 1200 m. The deeper geological body is number 5 with 1500 m. The deepest bodies include numbers 1 and 2 and the west of number 4 with a depth of 1700 m. We apply the cross-constraint joint inversion method in actual areas, and obtain the distribution of ore bodies used for geological interpretation.

V. Conclusion

In this article, we analyze the resolution of combined gravity joint inversion by eigenvalue spectrum. Combining simply model test, we infer the joint inversion of combined gravity has a higher resolution on the shallower space, and the airborne gravity highlights the deep resources. So we propose the cross-constraint joint inversion method for improving the vertical resolution of inversion results. A variety of model tests show the cross-constraint joint inversion method has a higher resolution than the traditional regularization joint inversion method, and the new method can improve the vertical resolution of deep space obviously. The airborne gravity data can be replaced by the upward continuation data, when airborne gravity data are observed at low altitudes. By the noise test, we can conclude that the cross-constraint joint inversion method has anti-noise performance. The cross-constraint joint inversion method is applied to the gravity data in Shaanxi province to get the high-density bodies, and the real gravity inversion results prove the practical applicability of the method.

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